

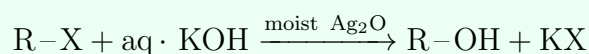
# Chapter 7: Alcohols, Phenols & Ethers

## Named Reactions Sheet + Tips & Tricks

### Formula Sheet

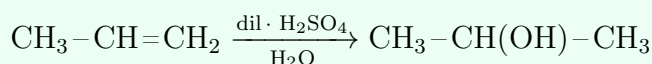
#### I. Preparation of Alcohols

##### From Alkyl Halides – Hydrolysis



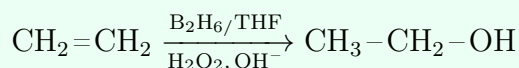
Nucleophilic substitution; aqueous KOH/NaOH or moist Ag<sub>2</sub>O/AgOH can be used.

##### Acid-Catalysed Direct Hydration of Alkenes (Markovnikov)



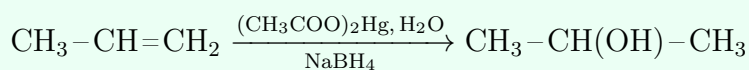
Follows Markovnikov's rule; uncatalysed addition of H<sub>2</sub>O to alkenes does not occur.

##### Hydroboration-Oxidation (Anti-Markovnikov)



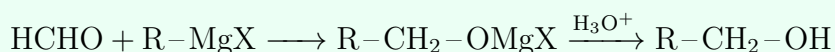
Indirect hydration; rate is inversely proportional to steric hindrance; BH<sub>3</sub> is unstable and exists as B<sub>2</sub>H<sub>6</sub>.

##### Oxymercuration-Demercuration (Markovnikov)



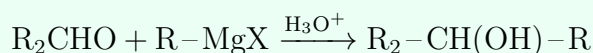
Addition of H<sub>2</sub>O following Markovnikov's rule, avoiding carbocation rearrangement.

##### From Grignard Reagent + Formaldehyde (1° Alcohol)

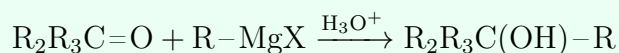


Formaldehyde gives a primary alcohol.

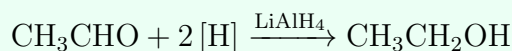
##### From Grignard Reagent + Aldehyde (2° Alcohol)



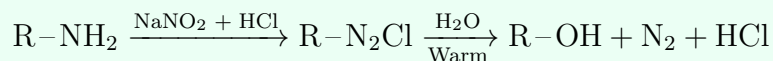
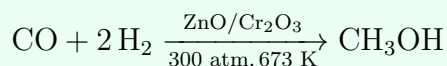
Other aldehydes give a secondary alcohol.

**From Grignard Reagent + Ketone (3° Alcohol)**

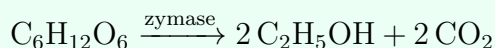
Ketones give a tertiary alcohol.

**Reduction of Carbonyl Compounds**

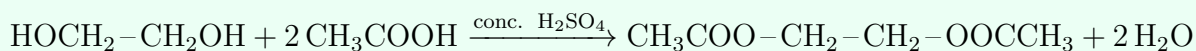
Reagents:  $H_2/Ni$ ,  $H_2/Pt$ ,  $H_2/Pd$ ,  $LiAlH_4$ , or  $NaBH_4$ . Aldehydes give 1° alcohols, ketones give 2° alcohols; 3° alcohols cannot be prepared this way.

**From Primary Amines****Methanol – Bosch Process**

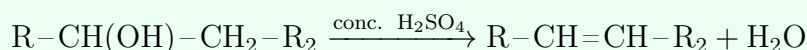
Industrial manufacture of methanol.

**Ethanol – Fermentation of Carbohydrates**

Starch is first hydrolysed:  $(C_6H_{10}O_5)_n \xrightarrow{\text{diastase}} \text{maltose} \xrightarrow{\text{maltase}} \text{glucose}$ , then glucose is fermented by zymase to ethanol.

**II. Chemical Properties of Alcohols****Esterification (Fischer Esterification)**

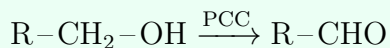
Mechanism: protonation of carbonyl O, nucleophilic attack by alcohol, proton transfer with loss of water, deprotonation gives neutral ester. General:  $R-COOH + R_2-OH \rightleftharpoons R-COOR_2 + H_2O$ .

**Dehydration of Alcohols (Saytzeff Rule)**

Mechanism: (1) protonation of OH, (2) loss of water to form carbocation, (3) loss of beta-H to form alkene; more substituted (major) alkene favoured.  $Al_2O_3/P_2O_5/POCl_3$  dehydrate without

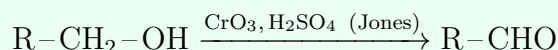
carbocation rearrangement (Saytzeff product only, no rearranged product).

#### Oxidation – PCC (Pyridinium Chlorochromate)



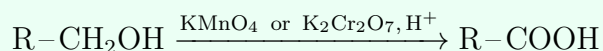
Oxidises 1° alcohol only to the aldehyde (does not go further to acid).

#### Oxidation – Jones/Collins/Sarett Reagents



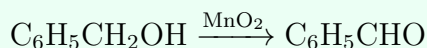
Jones reagent ( $\text{H}_2\text{CrO}_4$ ), Collins reagent ( $\text{CrO}_3 \cdot \text{pyridine}$  in  $\text{CH}_2\text{Cl}_2$ , 1:2:1), and Sarett reagent all selectively oxidise 1° alcohols to aldehydes.

#### Oxidation – Acidic $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ (Full Oxidation)



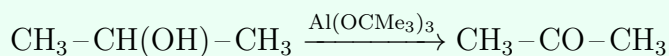
1° alcohol  $\rightarrow$  carboxylic acid; 2° alcohol  $\rightarrow$  ketone; 3° alcohol  $\rightarrow$  no reaction (only dehydrates to alkene).

#### Oxidation – $\text{MnO}_2$ (Allylic/Benzylic only)



Selectively oxidises allylic/benzylic alcohols only; here benzyl alcohol to benzaldehyde.

#### Oppenauer Oxidation (2° Alcohols only)

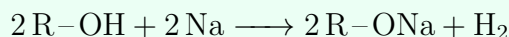


Only works on secondary alcohols, giving a ketone.

#### Reduction of Carboxylic Acids/Esters/Acid Chlorides – $\text{LiAlH}_4$



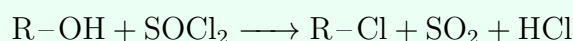
$\text{LiAlH}_4$  reduces all three (acids, esters, acid chlorides) to the primary alcohol;  $\text{NaBH}_4$  reduces none of these (reserved for aldehydes/ketones).

**Reaction with Sodium**

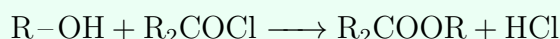
Forms sodium alkoxide; brisk effervescence observed.

**Reaction with Hydrogen Halides**

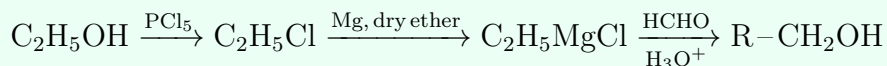
Reactivity of HX:  $\text{HI} > \text{HBr} > \text{HCl}$ . Reactivity of alcohol:  $3^\circ > 2^\circ > 1^\circ$ .

**Reaction with  $\text{PCl}_3$ ,  $\text{PCl}_5$ ,  $\text{SOCl}_2$** 

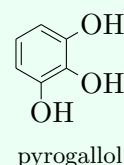
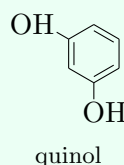
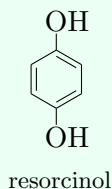
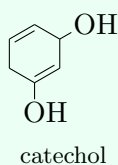
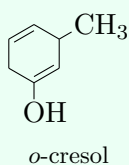
$\text{PCl}_3$ :  $3\text{R-OH} + \text{PCl}_3 \rightarrow 3\text{R-Cl} + \text{H}_3\text{PO}_3$ ;  $\text{PCl}_5$ :  $\text{R-OH} + \text{PCl}_5 \rightarrow \text{R-Cl} + \text{POCl}_3 + \text{HCl}$ ;  $\text{SOCl}_2$  is the Darzens process.

**Reaction with Acid Chlorides/Anhydrides**

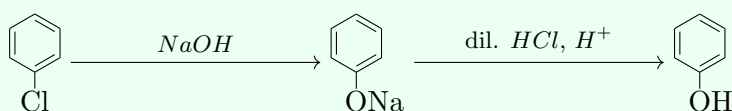
Forms ester; with anhydride:  $\text{R-OH} + (\text{R}_2\text{CO})_2\text{O} \rightarrow \text{R}_2\text{COOR} + \text{R}_2\text{COOH}$ .

**Conversion of One Alcohol to Another**

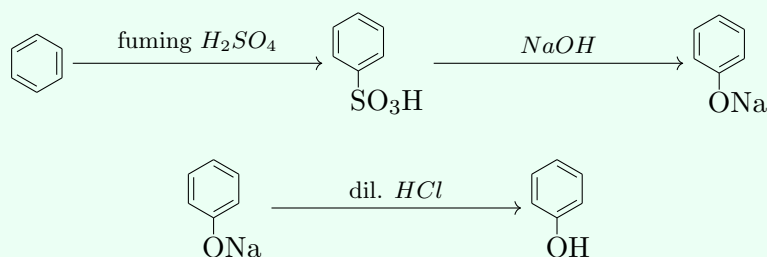
General route: alcohol  $\rightarrow$  alkyl halide  $\rightarrow$  Grignard reagent  $\rightarrow$  react with  $\text{HCHO}$ /aldehyde/ketone for  $1^\circ/2^\circ/3^\circ$  alcohol.

**III. Preparation of Phenols****Common Polyhydric Phenols**

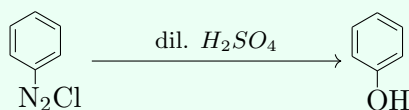
*o*-cresol = 2-methylphenol; catechol = 1,2-; resorcinol = 1,3-; quinol (hydroquinone) = 1,4-dihydroxybenzene; pyrogallol = 1,2,3-trihydroxybenzene.

**From Haloarenes – Dow's Process**

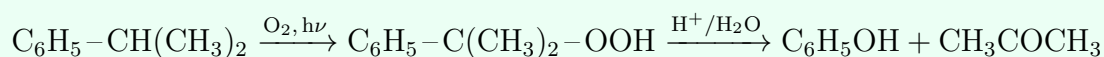
Chlorobenzene  $\rightarrow$  sodium phenoxide  $\rightarrow$  phenol.

**From Benzene Sulphonic Acid**

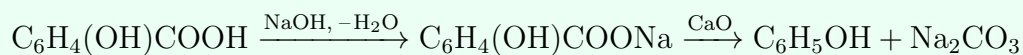
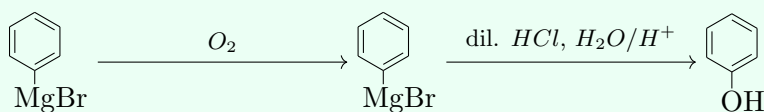
Benzene sulphonated, converted to sodium salt, then acidified to phenol.

**From Diazonium Salt**

Benzenediazonium chloride hydrolysed to phenol.

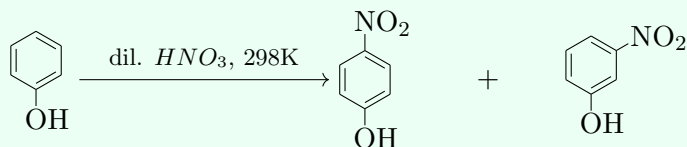
**From Cumene – Industrial Process (Hock Rearrangement)**

Step 1: benzene + propene  $\rightarrow$  cumene ( $\text{H}_3\text{PO}_4$ ). Step 2: cumene oxidised to cumene hydroperoxide. Step 3: acid-catalysed cleavage gives phenol + acetone. Major industrial process for phenol.

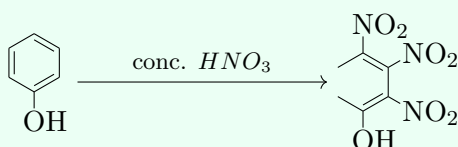
**By Decarboxylation of Salicylic Acid****From Grignard's Reagent**

Via phenylmagnesium bromide +  $\text{O}_2$ /hydrolysis  $\rightarrow$  phenol +  $\text{Mg}(\text{OH})\text{Br}$ .

## IV. Chemical Properties of Phenol

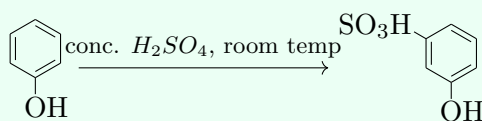
Nitration – Dilute  $\text{HNO}_3$ 

The *o*- and *p*-nitrophenol isomers are separated by steam distillation.

Nitration – Concentrated  $\text{HNO}_3$  (Picric Acid)

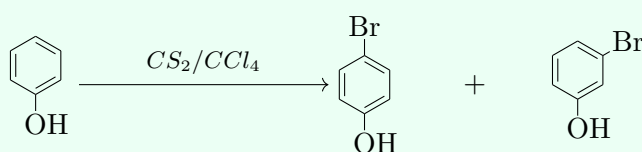
Forms 2,4,6-trinitrophenol (picric acid).

## Sulphonation



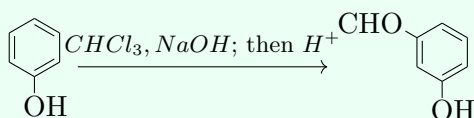
At room temperature gives ortho product; at 373 K gives the para product.

## Halogenation (Dilute, Less Polar Solvent)

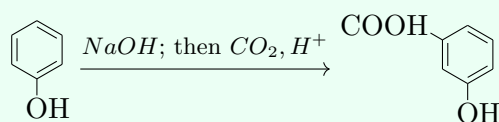


With excess  $\text{Br}_2$  (no solvent control), gives 2,4,6-tribromophenol directly.

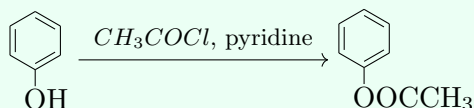
## Reimer-Tiemann Reaction



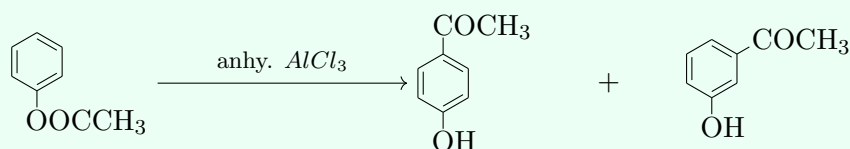
Phenol to salicylaldehyde; applied to anisole gives *p*-anisaldehyde.

**Kolbe's Reaction**

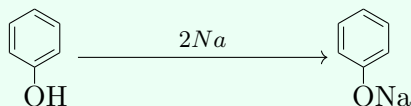
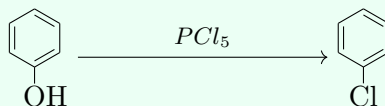
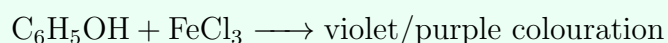
Phenol to salicylic acid via sodium phenoxide and  $\text{CO}_2$ .

**Esterification – with Acid Halide**

Forms phenyl ethanoate (phenyl acetate).

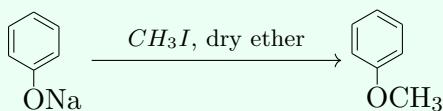
**Fries Rearrangement**

Rearranges phenyl ester to hydroxyacetophenone isomers.

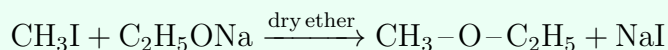
**Reaction with Sodium****Reaction with  $\text{PCl}_5$** **Test for Phenol –  $\text{FeCl}_3$  Test**

Neutral  $\text{FeCl}_3$  gives characteristic colouration with phenols.

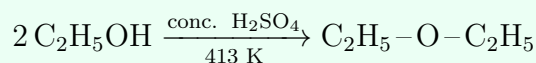
**V. Preparation and Properties of Ethers**

**Williamson's Synthesis – Anisole (Aryl Ether)**

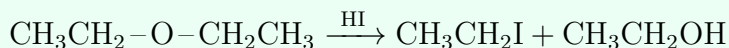
Sodium phenoxide + methyl iodide gives anisole.

**Williamson's Synthesis – Aliphatic Ether**

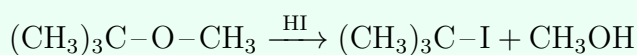
Alkyl halide must be 1° for a clean  $\text{S}_\text{N}2$  reaction (with 3° halides, elimination competes: e.g.,  $\text{CH}_3\text{--C}(\text{CH}_3)_2\text{--I} + \text{CH}_3\text{ONa} \rightarrow \text{CH}_3\text{--C}(\text{CH}_3)=\text{CH}_2 + \text{CH}_3\text{OH} + \text{NaI}$ ).

**Dehydration of Alcohols (Symmetrical Ether)**

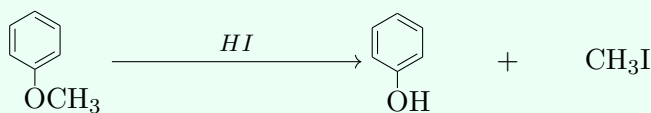
Gives diethyl ether (ethoxyethane).

**Cleavage by HI – Both Groups 1° ( $\text{S}_\text{N}2$ )**

Both alkyl groups primary: reaction proceeds by  $\text{S}_\text{N}2$ .

**Cleavage by HI – One Group 3° ( $\text{S}_\text{N}1$ )**

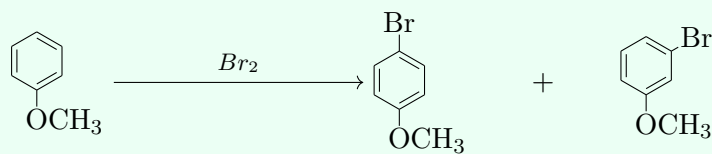
One alkyl group tertiary: proceeds by  $\text{S}_\text{N}1$ .

**Cleavage of Aryl Ether by HI**

With excess HI, phenol itself is not cleaved further (aryl-O bond resists cleavage); only the alkyl-O bond breaks.

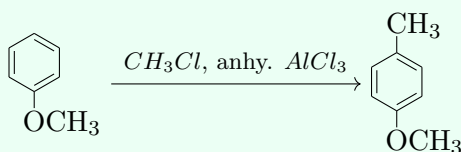


## Electrophilic Substitution of Anisole – Bromination

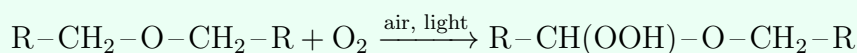


$-\text{OCH}_3$  is *o*, *p*-directing and activating.

## Friedel-Crafts Alkylation of Anisole



## Oxidation (Peroxide Formation)



Ethers slowly autoxidise on standing to form explosive peroxides.

Reaction with  $\text{PCl}_5$ 

## Tips, Tricks &amp; Hacks

- **Acidity order of substituted phenols:** electron-withdrawing groups (like  $-\text{NO}_2$ ) increase acidity (*p*-nitrophenol > *o*-nitrophenol > *m*-nitrophenol); electron-donating groups (like  $-\text{CH}_3$ ) decrease acidity (phenol > *m*-cresol > *p*-cresol > *o*-cresol).
- **Distinguishing 1°/2°/3° alcohols in a "difference" table:** boiling point highest for 1°, lowest for 3°; oxidation gives carboxylic acid (1°), ketone (2°), no reaction (3°); dehydration ease and  $\text{S}_{\text{N}}1$  substitution rate is fastest for 3°, slowest for 1° – opposite of  $\text{S}_{\text{N}}2$  rate.
- **Grignard + carbonyl shortcut:** formaldehyde  $\rightarrow$  1° alcohol; any other aldehyde  $\rightarrow$  2° alcohol; ketone  $\rightarrow$  3° alcohol – this single rule covers most Grignard-alcohol synthesis questions.
- **Fischer esterification** is reversible; use excess alcohol or acid, or remove water, to push equilibrium toward the ester (compare with Kolbe's electrolytic method – different topic, don't confuse with Kolbe's reaction for salicylic acid above).
- Carbocation rearrangement during acid-catalysed dehydration follows 1,2-hydride or 1,2-methyl shifts to reach the more stable carbocation *unless* the dehydrating agent is  $\text{Al}_2\text{O}_3$ ,  $\text{P}_2\text{O}_5$ , or  $\text{POCl}_3$ , in which case no rearrangement occurs (Saytzeff product directly).